

B. Tech. (common to all disciplines)-I/II Semester**Contact Hours (L-T-P): 3-0-2**

DPH001B	APPLIED PHYSICS	Total Credits: 3
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Course Objectives:

1. Students will be able to demonstrate competency and profound understanding of physics relevant to the various streams of engineering and technology.
2. Students will be able better to understand and learn the concepts of Quantum Mechanics and its applications.
3. Students will be able acquire the knowledge of semiconductors, laser system to meet desired needs within realistic constraints such as health and safety, manufacturability.
4. The graduates will able to understand the applications of quantum optics through Holography and communication through optical fibers and application of optical technologies.

UNIT 1	Introduction to Quantum Mechanics: Development of Quantum Mechanics, Compton Scattering, Uncertainty's Principle, Wave Particle Duality, Phase and Group velocities, Wave Packet, - Physical significance and its properties, Operators, Expectation values. Schrödinger's time dependent and time independent equations. Applications: Motion of a particle in one-dimensional box and three-dimensional box, Concept of Non-degeneracy and Degeneracy. Qualitative: Quantum Tunnelling (Barrier penetration) - Alpha Decay .
UNIT 2	Free Electron Theory and Band Theory of Solids: Overview of Drude & Lorentz theory, Quantum (Sommerfield's) theory of free electrons, Density of energy states, Fermi-Dirac Statistics, Fermi level. Band Theory of solids: Formations of band, Classification of Solids - Energy Band Diagram.
UNIT 3	Semiconductor Devices: Intrinsic and Extrinsic semiconductors, Carrier concentrations, Position of Fermi levels in semiconductors, Conductivity and Mobility, determination of bandgap, P-N junction diode: Formation, Energy band diagram, Calculation of internal potential barrier, diode equation and V-I characteristics under forward and reverse bias; Zener diode, Varacter diode, LED and Photodiode.
UNIT 4	Quantum Optics: Laser: Introduction to Coherence; Einstein's coefficients, relations between Einstein's coefficients, Threshold conditions for laser action. Solid-Nd-Yag Laser, Ruby laser, Gaseous-He-Ne, CO2 laser, and Semiconductor laser. Holography: Holography versus photography. Construction and re-construction of a Hologram and Applications. Communication: Optical fiber-Construction, Numerical Aperture and Angle of Acceptance, fractional refractive index, types of optical fibres and applications
UNIT 5	Electromagnetic Waves Scalar and Vector fields, Theorems- Gauss', Stokes and Gradient; Maxwell's equations – Differential and Integral Form, Significance of Equations, Ampere's Law and correction; Wave equation and its solution for free space. Transverse nature of EM waves; Poynting vector and theorem, Applications in communications

Course Outcomes

Upon successful completion of this course, the student will be able to:

- CO1: Understand the fundamentals of quantum mechanics to analyze the quantum behavior of matter in its micro state and its applications.
- CO2: Analyze and apply quantum theory and quantum statistics in understanding the physics of materials.
- CO3: Acquire the experiential learning with demonstration of knowledge about semiconductors and devices role in Electronics.
- CO4: Acquire and demonstrate the knowledge about the fundamentals of Lasers and its importance in real life.
- CO5: Learn the basics and nature of EM radiations and role in communications.

Suggested Books

1. Arthur Beiser, **Perspectives in Modern Physics**, McGraw Hill International.
2. H. S. Mani and G. K. Mehta, **Modern Physics**, East-West Press.
3. H Malik and AK Singh, **Engineering Physics**, McGraw Hill Education.
4. A. K. Ghatak, **Optics**, Tata McGraw Hill.
5. D. K. Bhattacharya and A. Bhaskaran: **Engineering Physics**, Oxford University Press.
6. S. Mani Naidu, **Engineering Physics**, Pearson.
7. M.N. Avadhanulu, P.G. Kshirsagar, “**A Textbook of Engineering Physics**”, S. Chand Publications
8. S. O. Pillai, **Solid State Physics**, Wiley Eastern.

MAPPING COURSE OUTCOMES LEADING TO THE ACHIEVEMENT OF PROGRAM OUTCOMES AND PROGRAM SPECIFIC OUTCOMES:

Course Outcome	Program Outcome												Program Specific Outcome		
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO1	PSO2	PSO3
CO1	H	M	M	H	M	L						M	H	M	
CO2	H	H	L	H	M	M						L	H	M	L
CO3	H	H	M		M	M	L	M	H			M	M	H	L
CO4	H	L	M		H	M	L		L			M	M	H	M
CO5	H	L	H	M	H	L						L	M	H	M

H = Highly Related, M = Medium, L = Low.

B. Tech. (common to all disciplines)-I/II Semester**Contact Hours (L-T-P): 3-0-2**

DPH002B	APPLIED PHYSICS LAB	Total Credits: 1
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List of Experiments

Students are required to perform any ten experiments out of the following list of experiments.

1	Study the V-I characteristics of p-n junction diode under forward and reverse bias.
2	Study the V-I characteristics of a zener diode under forward and reverse bias.
3	To study the V-I characteristics of Solar Cell
4	To determine the Planck's constant "h" by measuring radiations.
5	To study the variation in resistance of a Semiconductor with temperature and to determine its energy bandgap.
6	Study the temperature dependence of resistivity of a semiconductor (Four probe method) and to determine bandgap of semiconductor materials (Si and Ge).
7	Study of various LC and LCR circuits
8	To determine Resolving power of Telescope.
9	To determine Dispersive Power of a Prism using Mercury light source and Spectrometer.
10	To determine the wavelength of prominent lines of Mercury by using plane Diffraction Grating and Spectrometer.
11	To measure Numerical Aperture of an Optical Fiber.
12	To determine the profile of He-Ne LASER beam.
13	To determine wavelength of Sodium light source using Newton's Rings experiment.
14	*To study shift in fringes in interference experiment using Michelson's interferometer

Course Outcomes-

While graduating, students of the Applied Physics Lab program would be able to:

CO1: Learn the concepts and understandings through the experiments of electrical, electronics, optical and mechanical in engineering disciplines.

CO2: Develop the technical skill by conducting experiments and analyzing the reasons behind.

CO3: Demonstrate the concepts of Quantum optics via using modern tools for better learning.

CO4: Express their theoretical understanding more effectively by performing experiments and simulations.

CO5: Communicate their ideas effectively, both orally and in writing, and function effectively in multidisciplinary teams

MAPPING COURSE OUTCOMES LEADING TO THE ACHIEVEMENT OF PROGRAM OUTCOMES AND PROGRAM SPECIFIC OUTCOMES:

Course Outcomes	Program Outcome												Program Specific Outcome		
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	H	M	H		M	M						M	H	M	
CO2	H	M	L	M	L							M	H	L	
CO3	M	M	L		H	M						M	H	M	L
CO4		H		M	M			M			M	L	M	H	L
CO5		L						H	H	H		M	M	M	H

H = Highly Related; M = Medium L = Low